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Created 3 weeks ago

Z-Image AI: Free Fast S3-DiT Image Generator

Z-Image Turbo is a bit different from "classic" Stable Diffusion, so a lot of old prompting habits don't quite apply. I'll walk through how to prompt it deeply and safely, with special focus on *controlling* content (nudity, stereotypes, unwanted artifacts) even though the model **does not support traditional negative prompts at all.** (Hugging Face)

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1. How Z-Image Turbo thinks (and why "negative" prompts don't work)

Key facts that matter for prompting:

- It's a **6B single-stream diffusion transformer** (S3-DiT): text tokens and image tokens are processed together in one sequence. (GitHub)
- It's a **few-step distilled "Turbo" model** (about 8 actual diffusion steps), optimized for **fast, instruction-following generation**. (Hugging Face)
- It's trained for **bilingual prompts and text rendering (English + Chinese)** and follows written instructions unusually well. (Hugging Face)
- The Turbo model **does not use classifier-free guidance at inference**, so:
  - `guidance_scale` is usually set to **0**.
  - The model **ignores** `negative_prompt` entirely in the official pipeline. (Hugging Face)

> "Unlike traditional diffusion models, this model does **not** use negative prompts at all." – Z-Image authors (HuggingFace discussion) (Hugging Face)

So when your UI shows a "negative prompt" box, for Z-Image Turbo's official pipeline that box is *cosplay*—it won't change the image unless the UI is doing some custom hack.

Implication: You control nudity, style, artifacts, etc almost entirely via:

1. **Positive prompt constraints** ("wearing a tailored suit, fully clothed, modest outfit, no logos, plain background").
2. **Upstream / downstream filters** (safety classifiers, NSFW filters, manual review).
3. **Token choices** that avoid "baggage" (biased or over-stylized tokens).

Think of Z-Image Turbo as a very obedient camera crew + art director:

> If you don't say it, it's allowed. If you say it vaguely, it will improvise.

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2. Core positive-prompt structure for Z-Image Turbo

The Z-Image team recommends **long, detailed prompts**, and community testing has found that **camera-style, structured prompts** work best. (Hugging Face)

A good, reusable scaffold:

> [Shot & subject] + [Age & appearance] + [Clothing & modesty] + [Environment/background] + [Lighting] + [Mood] + [Style/medium] + [Technical notes] + [Safety/cleanup constraints]

Example (safe portrait):

> *A medium-shot portrait of an adult woman in her 30s, natural look, medium-length brown hair, **wearing a dark business suit and shirt, fully clothed, modest professional outfit**, standing in a modern office with a soft blurred background, soft diffused daylight, calm confident expression, realistic photography, 50mm lens, shallow depth of field, 4K quality, **plain background, no logos, no text, no watermark**.*

Why this works well:

- "Adult" + non-sexual context → steers away from sexualized depictions.
- Clothing is explicit ("business suit", "fully clothed", "modest").
- Background and artifacts are constrained ("plain background, no logos, no text, no watermark").
- Camera & lighting are clear, which Z-Image responds strongly to. (Z-Image Turbo)

Recommended sections to think through

Borrowing from a popular Z-Image best-practices article: (Z-Image Turbo)

1. **Composition**
   - Shot type: `close-up`, `medium shot`, `full-body`, `wide shot`.
   - Angle: `front view`, `45° angle`, `profile`, `looking slightly up`, `looking slightly down`.
2. **Character / subject**
   - Who they are: "adult software engineer", "elderly man", "group of office workers".
   - 2–4 traits: hair, build, expression, posture.
3. **Clothing + color palette**
   - 3–5 words is usually enough: `white oversized hoodie`, `dark blazer`, `simple blue dress`, `sportswear`, `formal suit`.
   - Palette: `warm palette`, `cool tones`, `muted colors`, `neon palette`.
4. **Environment / background**
   - Simple usually works best: `plain studio background`, `minimal interior`, `soft blurred city at night`.
5. **Lighting**
   - Z-Image responds *very* well to lighting keywords: `soft diffused daylight`, `cinematic warm key light`, `noir high-contrast lighting`, `studio portrait lighting`, `rim lighting`. (Z-Image Turbo)
6. **Mood / vibe**
   - `calm and professional`, `hopeful`, `dramatic`, `cozy`, `tense cinematic atmosphere`.
7. **Style / medium**
   - `realistic photograph`, `flat vector illustration`, `watercolor painting`, `manga page`, `pixel art`.

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3. Prompt length & parameters (just enough to matter)

3.1 How long should the prompt be?

- Z-Image Turbo actually likes **fairly long, detailed prompts** (hundreds of words are fine).
- In the official code, the default max text length is **512 tokens**; they suggest raising to 1024 tokens if you really need huge prompts. (Hugging Face)
- Practically: **80–250 words** of clear, structured description is a sweet spot.
  - Long *and precise* = good.
  - Long and "novel-y" / poetic = often worse.

You *can* run your "brief idea" through an LLM prompt-enhancer first (the Z-Image team mentions their own Prompt Enhancer script), but always **read and edit the result** to:

- Remove unnecessary flourish.
- Add your safety/constraints: clothing, age, no logos, etc. (Hugging Face)

3.2 Guidance / CFG & negative prompts

- Official Z-Image Turbo pipeline uses `guidance_scale = 0.0` for best quality. (Hugging Face)
- They explicitly state **negative prompts aren't used** for Turbo. (Hugging Face)
- Some third-party workflows (especially custom ComfyUI graphs) may re-introduce guidance, but unless you *know* that's happening, assume:
  - **CFG sliders above 0 won't help much.**
  - The "negative prompt" UI box is ignored by the base model.

So: **put all important constraints in the positive prompt**.

3.3 Seed, steps, resolution (practical defaults)

These aren't UI-specific, just general knobs:

- **Steps**: the official example uses 9 (8 effective). Start around **8–12 steps** and only go higher if you see obvious noise; Turbo is designed to work well with very few steps. (Hugging Face)
- **Resolution**: 1024×1024 is the "native" demo size; 768 or 512 can be fine for drafts. (Hugging Face)
- **Seed**:
  - Fix a seed while iterating your prompt → you see *prompt* differences instead of random noise differences. (Apatero Blog)
  - Randomize seed when you're exploring variety.

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4. "Negative" prompting *without* negative_prompt

Since Z-Image Turbo ignores negative prompts, we use **in-prompt constraints** instead. The goal is to say both:

- What **must** be there.
- What **must not** be there, encoded as positive constraints or "without/avoid" phrases.

4.1 General pattern

You can append a constraint clause to your prompt:

> "… realistic photography… plain studio background, no text, no watermark, no logos, no extra people in the background."

Even though the model isn't doing classifier-free guidance, modern text encoders still learn that phrases like "no watermark" or "without text" carry a meaning, and Turbo's strong instruction-following tends to respect them. (Tencent Cloud)

Useful patterns:

- `no text, no watermark, no logos`
- `plain background, not busy or cluttered`
- `no extra limbs, correct human anatomy`
- `no motion blur, sharp focus`
- `no lens distortion, no fisheye effect`

Add these near the **end** of the prompt, after you've established the main scene.

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5. Controlling nudity & sexualization (safely)

You specifically want to prevent nudity and unwanted "spice" the model might add. You can't rely on negative prompts, so:

5.1 Always specify:

1. **Age & role**
   - `adult man`, `adult woman`, `adult couple`, `group of adults`.
   - Put the word "**adult**" right next to human subjects to reduce ambiguity.
2. **Clothing & coverage**
   - Be explicit and boring in a good way:
     - `wearing a modest business suit`
     - `wearing a long-sleeved shirt and trousers`
     - `wearing casual jeans and a hoodie`
     - `fully clothed, everyday outfit`
   - If you care about coverage:
     - `arms and legs covered`, `no exposed midriff`, `no revealing clothing`.
3. **Context**
   - Everyday, non-sexual contexts: `office`, `street scene`, `family living room`, `classroom`, `conference stage`.
4. **Safety clause at the end**
   - `safe for work, non-sexual, fully clothed characters, no nudity, no suggestive poses.`

Example:

> *A full-body photo of an adult man and adult woman walking together on a city street, casual weekend scene, both **wearing jeans and light jackets, fully clothed, modest everyday outfits**, natural smiles, candid street photography, soft afternoon light, shallow depth of field, **safe for work, non-sexual, no nudity, no revealing clothing, no suggestive poses, no logos or text, plain blurred background**.*

This uses multiple overlapping signals:

- Clothing explicitly defined.
- "Fully clothed" + "modest" + "non-sexual" + "no nudity".
- Ordinary activity & setting.

5.2 Extra guardrails

Because Turbo is uncensored in many community builds, you may want *non-prompt* safety, for example:

- Automatic **image moderation** after generation.
- Simple pre-filter: reject prompts containing disallowed terms.
- Manual review in sensitive applications.

Those are outside prompting itself, but they're the most reliable layer for safety-critical use.

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6. Removing "baggage" from tokens (stereotypes & unwanted traits)

Tokens like "CEO", "witch", "rock star", "fashion model" often bring along **unwanted default looks** (gender, body type, makeup, etc.). You can:

6.1 Override with explicit traits

Example: you don't want "CEO" to default to a white male in a navy suit.

> *A professional group portrait of **four adult colleagues of diverse ethnicities and genders**, standing together in a modern office, smart-casual outfits instead of suits, friendly but professional expressions, equal framing, respectful and realistic depiction, safe for work, no exaggerated features, no stereotypes.*

Patterns:

- `diverse group of colleagues of different ages and ethnicities`
- `realistic body types, no exaggerated proportions`
- `light natural makeup` or `without makeup`
- `clean-shaven` / `short beard` / `no facial hair`
- `no tattoos` or `visible tattoos on one arm only` if you want control

6.2 Swap loaded tokens for neutral ones

Instead of:

- `businessman` → use `office worker`, `professional`, `team lead`.
- `sexy outfit` (don't) → `formal evening dress`, `modest cocktail dress`.

6.3 Role + description instead of single labels

Z-Image responds very well to "**role + 2–3 traits**":

> `a software developer, adult woman, short dark hair, glasses, wearing a hoodie and jeans, focused expression, working at a laptop`

This is more controllable than just "programmer".

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7. Using "negative-style" constraints to fix quality issues

Traditional negative prompts are often used to remove artifacts: extra fingers, blur, weird eyes, etc. With Turbo, embed that logic into the positive prompt:

Common issues & phrasing:

1. **Hands & limbs**
   - Add: `correct human anatomy`, `natural hands and fingers`, `no extra limbs or fingers`.
2. **Blur / noise**
   - Add: `sharp focus on the subject`, `clean detailed image`, `no motion blur, no grainy noise`.
3. **Weird background clutter**
   - Add: `simple, uncluttered background`, `nothing distracting behind the subject`.
4. **Logos / watermarks / stray text**
   - Add: `no text, no UI elements, no watermark, no branding, no company logos`.

Even general diffusion tutorials emphasize that the model does learn "avoid X" semantics from negative-ish phrasing; you're just putting that in your *one* prompt instead of a dedicated negative field. (Tencent Cloud)

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8. Bilingual prompting & text in images

Z-Image Turbo is unusually good at **English + Chinese text rendering and instruction following.** (Hugging Face)

Tips:

- Keep each chunk of text in **one language**:
  - Good: `a movie poster with the title "THE QUIET CITY" at the top in bold English letters`
  - Good: `Chinese title text "回忆之味" in large characters at the top`
- Describe **placement & style**:
  - `large white title at the top`,
  - `small subtitle line below`,
  - `Chinese vertical text along the right side`.
- For safety:
  - `no additional text except the title`,
  - `no random numbers or watermarks`.

Example bilingual poster prompt:

> *A minimalist movie poster, dark blue background with subtle city skyline silhouette at the bottom, **big English title text "QUIET STREETS" centered near the top**, small Chinese subtitle text "静谧之城" below it, clean modern typography, simple layout, no extra text, no logos, no watermark, safe for work.*

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9. Practical prompt templates (drop-in & safe)

You can adapt these to your workflow. They're written to be **safe, clothed, and artifact-controlled** by default.

9.1 Photorealistic professional headshot

> *A close-up headshot of an adult woman in her 30s, friendly confident expression, medium-length dark hair, **wearing a simple dark blazer over a light shirt**, studio portrait setting, soft diffused daylight from the front, subtle blurred gray background, realistic photography, 85mm lens, shallow depth of field, detailed but natural skin, **fully clothed, modest professional outfit, no jewelry except a small necklace, no logos, no text, no watermark, safe for work**.*

9.2 Group shot with diversity & no stereotypes

> *A medium-wide photo of **five adult coworkers of diverse ethnicities and genders** standing together in an open-plan office, casual smart outfits, laptops and desks in the background, soft natural daylight from large windows, relaxed but professional mood, realistic photography, **realistic body types, respectful and non-sexual depiction, fully clothed, no revealing clothing, no stereotypes, no logos, no text, clean unobtrusive background, safe for work**.*

9.3 Flat vector illustration (non-realistic style, same safety)

> *A flat vector illustration of an adult woman riding a bicycle through a park, stylized trees and simple buildings in the background, limited pastel color palette, clean modern design, no outlines, minimal shading, **modest casual clothing, long-sleeved shirt and trousers, fully clothed**, cheerful mood, clear blue sky, **no text, no watermark, no logos, simple uncluttered background**.*

9.4 Manga / comic page layout

> *A black-and-white manga page with four panels, clean inked lines, each panel showing an adult heroine exploring a futuristic city, expressive faces but non-sexualized, modest sci-fi outfit covering arms and legs, dynamic but readable compositions, Japanese sound effects integrated into the scenery, **no gore, no nudity, no suggestive poses, safe for work, clear panel borders, no watermarks or logos**.*

9.5 Product shot

> *A studio product photo of a modern wireless headset placed on a white table, soft shadow, subtle reflection, bright neutral background, soft box lighting from top left, realistic photography, 4K resolution look, extremely sharp details, **no hands, no people, no text, no logos besides a simple generic symbol on the headset, no watermark, plain white background**.*

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10. Quick checklist before you hit "Generate"

When you're about to send a prompt to Z-Image Turbo, ask:

1. **Did I explicitly say "adult" for all human subjects?**
2. **Did I describe clothing clearly and modestly?**
3. **Did I include a "safe for work / no nudity / non-sexual" clause when needed?**
4. **Did I specify background simplicity and "no logos / no watermark / no extra text"?**
5. **Did I define angle, shot type, lighting, and mood clearly?**
6. **Is any "negative prompt" box in my UI actually doing anything for Turbo? (Often: no.)**

If those are all "yes", you're using Z-Image Turbo in the way it was designed: **long, structured, camera-style prompts with explicit safety and quality constraints baked in**, instead of relying on old-school negative prompts that Turbo simply doesn't support.

If you'd like, next we can design a few domain-specific prompt templates (e.g., e-commerce, game concept art, training data generation) tailored to how you're using Z-Image Turbo.

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